



Neuralink demonstrated

Elon Musk demonstrated a working Neuralink brain implant in a live pig named Gertrude.
<http://digit.in/sep20-09>



Japanese flying car

Watch the successful model test flight of Japan's SkyDrive Inc's 'flying car'.
<http://digit.in/sep20-10>

Intel Core i9-10900K

Make way for the hot head

Intel's Comet Lake processors are based on the same Skylake architecture that came out in 2015. Over the years, Intel has made a few improvements to the process node and the IGP while continuously adding security mitigations to all the bugs that have cropped up over these years. With Comet Lake, the key improvements are the two additional cores, improved Turbo speeds with the introduction of Thermal Velocity Boost and a slightly improved IGP. Intel recently opened up about the delays to their 7nm process node and we all know how their 10nm process node has been faring. This isn't the last 14nm processor family from Intel, this will soon be followed by Rocket Lake which is also going to use the 14nm microarchitecture but we might finally see Intel's new Xe graphics in action. Rocket Lake is expected in Q4 2020 or Q1 2021 but until then, let's look at the 10th Gen Comet Lake processors with the Intel Core i9-10900K leading the way.

What's new?

The Intel Core i9-10900K boasts of two additional cores over the Coffee Lake Intel Core i9-9900K. Intel still has the advantage in terms of clock speeds and that puts them at an advantage over the AMD processors in certain workloads. And while the IGP isn't so good at gaming, the processor itself is better than AMD at games when coupled with a good discrete graphics card.

Another introduction with Comet Lake is Thermal Velocity Boost (TVB), a boost clock which works akin to NVIDIA's GPU Boost that has been around since the GTX 600 series of graphics cards. When the temperature of the processor is below a certain threshold, the processor will boost the frequency of the core by an additional 100 Mhz. This can be a single core or all cores, depending on how the processor is



loaded. In the case of Comet Lake-S, this temperature threshold is 70 degrees celsius. The Intel Core i9-10900K is a 10 Core processor with a rated TDP of 125W which means it will run quite hot. So you'll need some really good cooling to get TVB to kick in and to properly benefit from it. The 10900K has a Single Core TVB Turbo Boost 3.0 Max of 5.3 GHz and an All Core TVB of 4.9 GHz. Let's see how this fares in our test.

Performance

We'll be comparing the Intel Core i9-10900K against the AMD Ryzen 9 3900XT since that's the best we have from AMD at the moment. The 3900XT is a 12 Core processor and costs about 48K which is significantly cheaper than the Core i9-10900K which sells for 64.8K. It's the other way around in International markets as the 10900K retails for \$488 and the 3900XT for \$499. We start off with Blender which is a rendering tool and the AMD Ryzen 3900XT is slightly ahead of the 10900K in the BMW CPU benchmark. In the POV-Ray benchmark, we see the 10900K maintain a healthy lead over the 3900XT. In 7-Zip, we see the 3900X scores significantly higher in compression as well as

decompression runs. WinRAR, on the other hand, puts the 10900K ahead by a few seconds.

In our video transcoding tests with Handbrake we see that the two processors are neck and neck in most testing profiles. In all our browser-based benchmarks, the Intel Core i9-10900K performs better than the Ryzen 9 3900XT. Then with Cinebench, we see that the two are neck and neck in single-threaded performance while the 3900XT takes the lead in multi-threaded performance owing to its higher core count.

Coming to gaming, we see that Intel still maintains a strong hold over this domain. From popular titles such as

GTA V, Shadow of the Tomb Raider,

Far Cry 5, Strange Brigade to the slightly offbeat ones like Ashes of the Singularity, the Intel Core i9-10900K is ahead of the 3900XT by a respectable margin.

The power consumption on the 10900K is off the charts. We heard a little

coil whine from our older PSU and had to switch in a higher wattage one to do away with the noise. When fully loaded, the 10900K draws about 290 Watts which is way higher than the 220-230 Watts that the 3900XT draws.

Verdict

While this is still the same old Skylake microarchitecture, Intel is ensuring that they retain the performance lead in tasks that rely on clock speed since competing on core counts is something Intel isn't capable of at the moment. The Intel Core i9-10900K is a great processor for gaming and day-to-day tasks but for the silly pricing they have in India, we don't see it selling well.

—Mithun Mohandas

SPECIFICATIONS

CLOCK SPEED: 3.7 GHz | TURBO FREQUENCY: 5.3 GHz | CORES/THREADS: 10/20 | LITHOGRAPHY: 14nm | TDP: 125W | CACHE: 20 MB Smartcache | PCIe LANES: 16 | PCIe VERSION: 3.0 | MEMORY SUPPORT: DDR4.

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